3.1 Functions

Name _____ Period ____ Date _____

Algebra I 3.1 · Student sheet

I can…

- determine whether a relation is a function.
- find the domain and range of a function from a graph or a real-life situation.
- identify the independent and dependent variables.

Vocabulary · fill in the missing word

Relation — a _____ of inputs with outputs

Function — a _____ that pairs each input with exactly one output

Domain — the set of all _____

Range — the set of all _____

Independent variable — the _____ variable (x)

Dependent variable — the _____ variable (y); its value depends on the input

Vertical Line Test — a graph is a _____ if no vertical line crosses it more than once

Part 1 · Function or not?

Relation	Function?	Why? (each input has …)
(1, 2), (2, 3), (3, 4)		
(1, 2), (1, 3)		
(−1, 5), (0, 5), (1, 5)		
student → locker number		
teacher → students		
person → birthday		

Part 2 · Notes and examples (fill in each note, then work the example)

Function: A function pairs each input with exactly _____ output.

Vertical Line Test: Vertical Line Test: if a vertical line crosses the graph _____, it is not a function.

Example 1

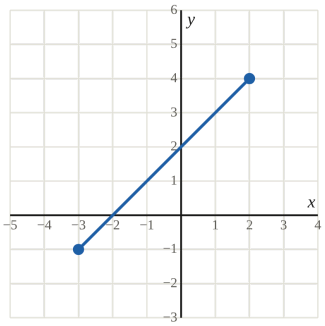
Is each relation a function? Explain.

a. (−2, 1), (0, 3), (2, 1), (4, 5) b. (1, 2), (3, 4), (1, 6)

Domain and range: Domain: the set of all _____. Range: the set of all _____.

Example 2

Find the domain and range of the function shown.



Example 3

The total cost C of T-shirts depends on the number n of T-shirts bought. Identify the independent and dependent variables.

Example 4

You work up to 4 hours and earn $10 per hour: P = 10h. Find the domain and range.

Part 3 · Partner practice

Partner 1

Is (5, 0), (6, 0), (7, 1) a function?

Partner 2

Find the range of $y = 2x + 1$ for the domain $-1, 0, 1, 2$.